Jinwoo Kim

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EDUCATION

Mar 2016 — Present

Korea Advanced Institute of Science and Technology

Daejeon, South Korea

B.S. Bio and Brain Engineering, B.S. Computer Science

Cumulative GPA 4.07/4.3
Relevant Coursework: Artificial Intelligence, System Programming, Algorithm, Data Structure, Discrete Mathematics, Probability & Statistics, Linear Algebra, Differential Equations, Multivariate Calculus

EXPERIENCES

Jan 2020 — Present

Research Intern, Vision and Learning Laboratory

KAIST CS

- Conducting **computer vision** research on scene understanding, using **PyTorch** and **OpenCV**
- Applying **transformers** and **variational autoencoders** to model object layout in a scene
- Using **Docker** for contained testing in Linux server environment

Aug 2018 — Dec 2019

Research Intern, Visual System Neural Network Laboratory

KAIST BiS

- [**Journal paper under review**] **Kim J***, Song M* & Paik SB (2020). *Spontaneous retinal waves generate long-range horizontal connectivity in visual cortex*. BioRxiv.
- Developed an **unsupervised neural network** model that mimics the development of visual functional circuits from retinal topology, using **MATLAB**.
- Carried out **simulation, statistical analysis,** and **academic writing** as a co-first author.
- Presented a **conference poster** at Society for Neuroscience 2019, Chicago.

Sep 2019 — Nov 2019

Naver AI Hackathon

- Ranked 22/100 in automated speech recognition competition.
- Implemented state-of-the-art **audio processing** and **deep learning** techniques on top of LSTM Seq2Seq, using **Python**, **Librosa**, and **PyTorch**

INDEPENDENT PROJECTS

Sep 2019 — Nov 2019

Zero-Shot Korean Voice Cloning

- Applied **autoencoder** and task-driven speaker embedding for **zero-shot learning** of voice cloning from 5 seconds of the individual speech sample.

Apr 2020 — May 2020

DNN Acceleration with CUDA

- Sped up CNN layers by offloading computations to GPU with low-level **CUDA programming**
- Achieved 2.5x acceleration of YOLOv2-tiny object detector over NumPy

HONORS

Mar 2018 — Present

National Science & Technology Scholarship

Feb 2018

Dean's List, School of Engineering KAIST

Mar 2017 — Present

Alumni Scholarship

SKILLS

Experienced: Python, C/C++, Numpy, PyTorch, Matlab, Git, Linux

Skillful: R, Tensorflow, CUDA, SQL, AWS, LaTeX

C1 level English